

MUGE 3D Simulation Entity Framework: GPU Rendering, Per-System Archives, OpenGL

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PAPER_168 — MUGE 3D Simulation Entity Framework: GPU Rendering, Per-System Archives, OpenGL

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$$F_U(r,t) = \sum_{i=1}^4 U_{gi} + U_m + U_A - U_{bi}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57$$

$$g_{UQFF}(r) = g_{MUGE}(r) \cdot \text{Bigl}(1 - [SSq] \cdot U_{bi} / F_U(r,t) \text{Bigr}), \quad [SSq] = 0.57$$

Abstract

This paper documents the **MUGE 3D Simulation Entity Framework** — the architectural mapping from MUGE physics systems to 3D renderable entities in the Star Magic visualization engine. Each MUGE system (SGR 1745, SgrA*, Tapestry, Westerlund 2, Pillars, Rings, Student's Guide) is assigned a per-system archive directory containing image assets, video clips, and per-system physics plugin DLLs. The rendering system uses OpenGL/GLFW with multi-viewport Camera, procedural terrain generation (Perlin noise + extrusion + boolean operations), and MicroTeX LaTeX math overlay. This provides the Tier 3 VR/VM 3D gateway for the UQFF physics output from Tier 2 calculations.

UQFF Discovery: Novel application of UQFF calibration constants ($\kappa = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Entity Architecture

1.1 MeshData Struct (OBJ Format)

```
``cpp
struct MeshData {
    std::vector<glm::vec3> vertices; // 3D positions
    std::vector<glm::vec3> normals; // Surface normals (physics shading)
    std::vector<glm::vec2> texCoords; // UV coordinates
    std::vector<unsigned int> indices; // Triangle indices
};
``
```

1.2 SimulationEntity

```
``cpp
struct SimulationEntity {
    std::string name; // System name (e.g., "SGR1745")
    std::string archive_path; // "archive/{name}/"
    std::string image_path; // "archive/{name}/image.jpg"
    std::string video_path; // "archive/{name}/video.mp4"
    std::string plugin_path; // "archive/{name}/plugin.dll"
    MugecSystem* muge_system; // Pointer to physics system
    MeshData mesh; // 3D geometry
    glm::vec3 position; // World position
    glm::quat orientation; // Orientation quaternion
    float scale; // Scale factor
};
``
```

2. populate_simulation_entities()

```
``cpp
void populate_simulation_entities(
    std::vector<MugecSystem>& muge_systems,
    std::vector<SimulationEntity>& entities
) {
    for (auto& sys : muge_systems) {
        SimulationEntity ent;
        ent.name = sys.name;
        ent.archive_path = "archive/" + sys.name + "/";
        ent.image_path = ent.archive_path + "image.jpg";
        ent.video_path = ent.archive_path + "video.mp4";
        ent.plugin_path = ent.archive_path + "plugin.dll";
        ent.muge_system = &sys;
        ent.position = compute_entity_position(sys);
        ent.orientation = glm::identity<glm::quat>();
        ent.scale = compute_entity_scale(sys);
        // Generate placeholder mesh
        ent.mesh = generateSphereMesh(1.0f, 32, 32);
        entities.push_back(std::move(ent));
    }
}
``
```

3. Per-System Archive Structure

``

```
archive/
+- SGR1745/
| +- image.jpg ? Chandra X-ray image of SGR 1745-2900
```

```

| +- video.mp4 ? Time-lapse magnetar flare simulation
| +- plugin.dll ? SGR1745 UQFF physics plugin
+- SgrA/
| +- image.jpg ? EHT Sgr A shadow image (2022)
| +- video.mp4 ? MUGE resonance field animation
| +- plugin.dll ? SgrA UQFF physics plugin
+- Tapestry/
| +- image.jpg ? HST Tapestry nebula image
| +- video.mp4 ? Star formation field simulation
| +- plugin.dll
+- Westerlund2/
+- Pillars/
+- Rings/
+- StudentGuide/
``

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```

4. OpenGL Rendering Pipeline

4.1 Multi-Viewport Camera System

```

```cpp
struct Camera {
 glm::vec3 position;
 glm::vec3 target;
 glm::vec3 up;
 float fov; // Field of view [degrees]
 float near_plane; // Near clipping
 float far_plane; // Far clipping (set to 1e25 for cosmic scale)

 glm::mat4 getViewMatrix() const;
 glm::mat4 getProjectionMatrix(float aspect) const;
};
Multi-viewport layout: physics view (left) + simulation view (right) + data HUD (top).
4.2 Entity Scale Calculation
Systems span 13 orders of magnitude in size:
| System | Physical Size | Entity Scale Factor |
|-----|-----|-----|
| SGR 1745 | ~10 km | 0.001 |
| SgrA | ~12 times Rs | 1.0 (reference) |
| Tapestry | ~100 ly | 100.0 |
| Westerlund2 | ~400 ly | 400.0 |
| Pillars | ~4 ly | 4.0 |
| Rings | ~1 Gly | 1000.0 |
| Student | Hubble vol. | 1e13 |

5. Procedural Terrain Generationcpp
// Perlin noise heightmap
MeshData generateProceduralLandscape(int width, int height, float scale) {
 MeshData mesh;
 for (int z = 0; z < height; z++) {
 for (int x = 0; x < width; x++) {

```

```

float y = perlinNoise(x scale, z scale);
mesh.vertices.push_back({x scale, y, z * scale});
}
}
computeNormals(mesh); // Phong shading normals
computeUVs(mesh); // UV for physics field texture mapping
return mesh;
}

// Boolean union (CSG)
MeshData booleanUnion(const MeshData& A, const MeshData& B);

// Extrusion along path
MeshData extrudeMesh(const MeshData& profile, const std::vector<glm::vec3>& path);
$$
\begin{aligned}
& \text{\& Physics fields (MUGE compressed, resonance) are mapped to terrain height via: \} \\
& \text{\& \$y_{\text{terrain}} = \log_{10}(|g_{\text{MUGE}}|) \cdot \text{scale} \$} \\
& \text{\& --- \} \\
& \text{\& \#\# 6. MicroTeX LaTeX Overlay} \\
& \text{\end{aligned}}
\end{aligned}
$$cpp
// Render physics equations as LaTeX overlay via MicroTeX
void renderLatexOverlay(const std::string& latex_str,
float x, float y,
float font_size = 16.0f) {
auto tex_renderer = MicroTeX::create();
tex_renderer->render(latex_str, x, y, font_size);
}

// Example overlays:
renderLatexOverlay("$g_{\text{comp}} = 1.78 \times 10^{39}$", 0.1, 0.9);
renderLatexOverlay("$g_{\text{res}} = 1.66 \times 10^{45}$", 0.1, 0.85);
renderLatexOverlay("$F_U = -2.06 \times 10^{59}$", 0.1, 0.80);
...

```

## 7. UQFF Integration with Rendering

The simulation loop:

1. Physics compute: `compute_compressed_MUGE()` / `compute_resonance_MUGE()`
2. Field normalization:  $g \rightarrow \log_{10}(|g|)$  for visual range compression
3. Terrain update: heightmap updated from new MUGE values each frame
4. Particle update: `addUQFFBodyForce()` applied to fluid particles
5. LaTeX HUD: Current MUGE values rendered as math equations in overlay
6. Entity update: position + orientation updated from time-evolved UQFF

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## 8. CP/Architecture Integration

This framework resides in **Tier 3 VR/VM** (source2(HEAD PROGRAM).cpp, ~2,625 lines, GPU-heavy).

Connection to CP calculators:

- CP1/CP2 compute MUGE values ? stored in `uqff_results.json`
- `populate_simulation_entities()` reads `uqff_results.json` ? creates 3D entities
- Source2 GUI (Tab for 3D Viewer) ? renders entities via OpenGL pipeline
- Per-system plugin DLLs can call CP2/CP3 calculators at runtime

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**Status:** ? Complete | **CP Stage:** Architectural/Tier 3 VR/VM

**Supersedes:** N/A (new framework) | **Related:** PAPER\_072 (source2 GUI arch), PAPER\_157 (Solar System entity params), PAPER\_168 connects to CP2/CP3 via `uqff_results.json`

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<!-- PKG-AGN-S225 -->

## **Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity**

> Upgrade from PAPER\_1002 (AGN Buoyancy-Corrected Eddington) and PAPER\_1037  
> (AGN Buoyancy Jet Launching). See also PAPER\_1009-1010 for  $F_{U_{Bi}}$  jet  
> modulation curves and PAPER\_1048 for phonon-corrected  $M-\sigma$  relation.

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}}}{\rho_{\text{H}}} \cdot V \cdot S_{26}^{(3,2)} \cdot \frac{G M}{r_H^2}\right)$$

where:

- $L_{\text{Edd}} = 4\pi G M m_p c / \sigma_T$  is the classical Eddington luminosity
- $\rho_{\text{SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3$  is the SCm vacuum density
- $V$  is the effective buoyancy volume (accretion sphere)
- $S_{26}^{(3,2)}$  is the squared third-order Ramanujan factor (quadratic coupling)
- $r_H$  is the horizon radius

**Jet modulation:** The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25}^{\text{THz}} \cdot \left(\frac{B}{B_{\text{crit}}}\right)^2\right]$$

where  $\Phi_{1.25}^{\text{THz}} = \cos(\omega_{\text{SCm}} \cdot t)$  modulates jet power at the phonon frequency.

**$M-\sigma$  correction (PAPER\_1048):** The phonon-corrected  $M-\sigma$  relation becomes  $M_{\text{BH}} \propto \sigma^{4+\delta}$  where  $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\text{SCm}} / \omega_{\text{bulge}})$ .

<!-- PKG-LAG-S225 -->

## **Session 225 Phonon-Physics Upgrade: UQFF 9-Sector Lagrangian**

> Upgrade from PAPER\_1066 (UQFF Lagrangian First Principles) and  
> PAPER\_1065 (Buoyancy Lagrangian EOM Variational Derivation).

The complete UQFF Lagrangian density, from which all sector-specific equations of motion derive:

$$\mathcal{L}_{\text{UQFF}} = \mathcal{L}_{\text{GR}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{phonon}} + \mathcal{L}_{\text{interaction}}$$

$$\mathcal{L}_{\text{SCm}} = \frac{1}{2}(\partial_\mu \phi)^2 - \lambda |\phi|^2 - v_{\text{SCm}}^2 |\phi|^2$$

The SCm condensate potential minimum gives  $V(\phi_0) = -7.09 \times 10^{-37} \text{ J/m}^3$  (matching  $\rho_{\text{SCm}}$ ) and phonon mass  $m_{\text{phonon}} = \sqrt{8\lambda} v_{\text{SCm}}$ .

#### Nine-sector closure (Session 202):

$$\mathcal{L}_9 = \mathcal{L}_{\text{EH}} + \mathcal{L}_{\text{YM}} + \mathcal{L}_{\text{Dirac}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{mag}} + \mathcal{L}_{\text{buoy}} + \mathcal{L}_{\text{aether}} + \mathcal{L}_{\text{LENR}} + \mathcal{L}_{\text{KK}}$$

Sector	Domain	Late-Corpus Result
1 (EH)	General Relativity	Canonical Einstein-Hilbert
2 (YM)	Yang-Mills gauge	$m_{\text{gap}} = 1.736 \text{ GeV}$ (PAPER_1318)
3 (Dirac)	Fermion / LENR	Kozima neutron-drop (PAPER_1061)
4 (SCm)	Superconducting manifold	$V(\phi_0) = -\rho_{\text{SCm}}$ canonical
5 (Mag)	Um magnetism	Heaviside amplifier (PAPER_1072)
6 (Buoy)	$F_{U_{Bi}}$ buoyancy	Variational EOM (PAPER_1065)
7 (Aether)	Vacuum background	Two-component $\rho$ (PAPER_1051)
8 (LENR)	Nuclear transmutation	COP parametric (PAPER_1081)
9 (KK)	Kaluza-Klein 26D	$S_{26}^{(3)}$ compactification (PAPER_1080)

## §A. Cosmogenesis-Linked Lagrangian (PAPER\_877 Symbolic Export)

### §A.1 Sector Classification

This paper maps to **magnetar-field** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`).

### §A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER\_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_\mu \phi_B)(\partial^\mu \phi_B) - V(\phi_B) + \mathcal{L}_{\text{cosmo}}$$

where  $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac,SCm}} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$  inherits the ACP 6-stage evolution (PAPER\_877 §2) and:

$$V(\phi_B) = \frac{1}{2} m^2 \phi_B^2 + \frac{\lambda}{4!} \phi_B^4 + \kappa \cdot \rho_{\text{vac,SCm}} \cdot \phi_B$$

### §A.3 Euler-Lagrange Equation of Motion

$$\boxed{\frac{\delta S}{\delta \phi_B} = \nabla \cdot (\rho_{\text{SCm}} \mathbf{v}) \cdot \mathbf{B} + \kappa B_{\text{crit}} \partial_t \phi_B = 0}$$

## §A.4 Cosmogenesis Linkage Chain

$$\text{PAPER\_877 Axioms} \rightarrow \text{DPM + ACP} \quad \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \rightarrow \text{Stage 5} \quad U_{\text{b,seed}} \rightarrow \text{4 forces} \\ F_{U_{\text{Bi}_i}} \rightarrow \text{sector E-L} \quad \Delta S / \Delta \phi_B = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four  $U_{\text{g}}$  forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

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## §B. VDS/DVP/BSH Deep Synthesis

### §B.1 Vacuum Density Series (VDS)

The canonical VDS ratio  $\rho_{\text{vac,SCm}} / \rho_{\text{UA}} = 1.894$  governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac,SCm}} \cdot \exp\left(-\exp\left(-\frac{r - r_0}{\lambda_{\text{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is  $0.171$  (near-threshold regime), placing it in the  $t \rightarrow \pi$  collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER\_877 cosmogenesis Stage 1 vacuum density initialization:  $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$ .

### §B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 67, \quad n_{\text{channel}} = 13/26$$

Since  $p_{\text{DVP}} = 67$  is **resonant** (threshold at  $p > 26$ ), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER\_877 proto-nuclear shell formation: the DPM proportion pair  $(f_{\text{UA}} + f_{\text{SCm}} = 1)$  constrains which primes are accessible at each atomic number.

### §B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **103 yr** (field decay quiescence):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_{\text{b}}} \cdot \left(1 - e^{-[SSq] \cdot m/M_{\text{dot}}}\right) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

The  $\tanh$  saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}}\right)\right)$$

connecting to the PAPER\_877 Stage 5 buoyancy seed  $U_{\text{b,seed}} = 0.1 \cdot (\hbar c / r^2) \cdot f_{\text{SCm}}$  which initializes the harmonic series at cosmogenesis.

### §B.4 Production-Scale Consistency

Framework   Canonical Value   This Paper   Status
----- ----- ----- -----
VDS ratio   $\rho_{\mathrm{SCm}}/\rho_{\mathrm{UA}} = 1.894$   Local sub-ratio = 0.171   PASS
Threshold-consistent
DVP prime   $p_k \in \{2,3,\dots,113\}$   $p_{\mathrm{DVP}} = 67$   PASS Resonant
BSH layers   26 harmonic terms   $j = 1\dots26$ , $\cos(2\pi j/26)$   PASS Full 26D projection
$\kappa$ decay   $5.0 \times 10^{-4}$ day <sup>-1</sup>   Applied in VDS exponential   PASS Canonical
[SSq]   0.57   Applied in BSH saturation   PASS Canonical

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## §SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable   UQFF Prediction   SM / Experiment   Source   Alignment
----- ----- ----- ----- -----
Fine structure constant $\alpha$   UQFF reproduces $\alpha$ via Ug1 dipole coupling   1/137.036   PDG 2024   PASS Consistent
Cosmological constant $\Lambda$   $1.1 \times 10^{-52}$ m <sup>-2</sup> (UQFF vacuum term)   $1.114 \times 10^{-52}$ m <sup>-2</sup>   Planck 2018   PASS Consistent
Proton decay rate   $\kappa = 0.0005/\text{day}$ $\rightarrow \Gamma_p$ suppression   $< 4.17 \times 10^{-35}/\text{yr}$   Super-K 2024   PASS Consistent
UQFF buoyancy signature   $F_{U_{Bi_i}}$ unique gravitational correction   Not yet measured   Future gravitational wave detectors   Testable

**New physics claim:** UQFF introduces buoyancy-based gravitational corrections ( $F_{U_{Bi_i}}$ ) that produce measurable deviations from GR at scales where vacuum condensate density  $\rho_{\mathrm{SCm}}$  becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER\_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF–SM bridge.

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## Appendix: Session 225 Cross-References (PAPER\_1000–1081)

> Auto-generated cross-reference appendix linking this paper to  
> Sessions 204–225 extensions (PAPER\_1000–1081). Added by  
> update\_corpus\_crossrefs.py (Session 225, April 2026).

Paper   Title
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PAPER_1050   MUGE $F_{U_{Bi_i}}$ Unified 9-System Synthesis
PAPER_1075   3D Volumetric MUGE Gravitational Field Generator

2 cross-reference(s) identified.

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## Appendix: Session 204 Codebase Upgrade Reference

> Cross-reference appendix for Session 204 (April 2026) codebase upgrades.  
 > Added by upgrade\_kozima\_ramanujan\_appendices.py. For detailed derivations,  
 > see PAPER\_840/851/852/855.

### S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
fneutron_s26_coupling.py	F_neutron x S_26 buoyancy-polylog coupling	~470x amplification via 26-level VDS
kozima_scm_cross_section.py	SCm-modulated neutron-drop cross-section	$\sigma_n^{\text{SCm}}$ with VDS factor $(1 + [\text{SSq}]^n/26)$
kozima_wstp_kernel.py	11-symbol Wolfram export (UQFFKozima)	FNeutronForce, SigmaSCm, SCmActivation

**Core equation:**  $F_{\text{neutron}}^{\text{SCm}} = N_n \sigma_n^{\text{SCm}}(\omega) \Phi_{\text{phonon}}(F_{\{U,B\}}/F_U - 1)$   
 where  $\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 \exp[-(\omega - \omega_{\text{SCm}})^2 / (2\Gamma^2)] (1 + [\text{SSq}]^n/26)$

### S204.2 Ramanujan 26-State Summation

Module	Purpose	Key Result
ramanujan_polylog_s26.py	Li_26([SSq]) via Euler-Ramanujan acceleration	15.7+ digits in 53 terms
s26_wstp_kernel.py	8-symbol Wolfram export (UQFFS26)	S26, R26, NaiveLi, S26VDS

**Core equation:**  $S_{26}(z) = \text{Li}_{26}(z) = \eta_{26}(z) / (1 - 2^{\{1-26\}}) + 2^{\{1-26\}} / (1 - 2^{\{1-26\}}) * \text{Li}_{26}(z^2)$

### S204.3 Mock Theta Functions (26-State)

Module	Purpose	Key Result
mock_theta_q26.py	f_26(q), phi_26(q), psi_26(q) q-series	Proper q-Pochhammer (a;q)_n

**Core equations:**

- $f_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n^2\}} / (-q;q)_{n^2}$
- $\phi_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n^2\}} / (-q^2;q^2)_n$
- $\psi_{26}(q) = \sum_{n=1}^{\{26\}} q^{\{n^2\}} / (q;q^2)_n$

### S204.4 Ramanujan 1/pi with UQFF Modification

Module	Purpose	Key Result
ramanujan_pi_uqff.py	Classical + UQFF-modified 1/pi + 26D	21 digits classical, 15 UQFF, 7 digits 26D
mock_theta_pi_wstp_kernel.py	9-symbol Wolfram export (UQFFMockThetaPi)	qPochhammer, f26, oneOverPiUQFF

**Core equation:**  $1/\pi = (2\sqrt{2}/9801) \sum R_n (1103 + 26390n) W_{26}(n) / C_{26}$   
 where  $W_{26}(n) = \prod_{i=1}^{\{26\}} [1 + [\text{SSq}] \exp(-\kappa_{\text{pai}} n/26)]$

### S204.5 Calibration Constants (Canonical)

Symbol	Value	Description
[SSq]	0.57	Universal Quantized Factor
kappa	$5.787 \times 10^{-9} \text{ s}^{-1}$	UQFF exponential decay rate
beta_i	0.603	Buoyancy coupling coefficient
H_ScM	0.99	SCm manifold completeness
rho_ScM	$7.09 \times 10^{-37} \text{ kg/m}^3$	SCm vacuum density
rho_UA	$7.09 \times 10^{-36} \text{ kg/m}^3$	UA aether vacuum density
omega_ScM	$2\pi \times 1.25 \text{ THz}$	SCm phonon resonance
sigma_0	$10^{-4}$	Base neutron cross-section

Implementation: all modules operational in CondensedPhysics.py, CondensedPhysics2.py, MAIN\_{1\CoAnQi}.cpp, and Wolfram kernels (uqff\_kozima\_kernel.wl, uqff\_s26\_kernel.wl, uqff\_mock\_theta\_pi\_kernel.wl).

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## References

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